Database Class Project

Prof. Rachlin / Introduction to Databases

Overview

The CS3200 class project is an opportunity to develop a database application or carry out a data science research project that showcases your own interests. There are a wide range of possibilities, but the expectation is that this project will have some real-world significance, demonstrate an innovative application, or provide insights into an integrated database. What is "significant"? If it is a project that you would be proud to discuss with a potential employer, then it is probably significant and an excellent class project. If it is a data integration project that demonstrates the ability to ask deeper scientific questions, that too is probably significant.

Good projects (as well as good commercial ideas and good research projects) are often based on an existing idea but apply some unique twist. Perhaps you could develop a novel dating website, a messaging platform, or a next-generation Instagram. Maybe you'd like to build a platform for a citizen science initiative that stores observations by amateur naturalists, a bioinformatics database that integrates information about genes, diseases, clinical trials, bioassays, and chemical compounds, or maybe some type of online store.

Always keep in the back of your mind that an interviewer will someday ask the question: *"Tell me some interesting projects you've worked on and why were they interesting to you?"* and you will want to have a good answer!

Milestones

The project has 3 milestones. (See Canvas for due dates):

1. **Project Proposal**: Describe the database application you intend to design and build and its significance. Identify all project members. (1 page PDF.)

2. **Final Project Implementation**: Document and deliver a fully functional application or data science research report. All tables should be in 3rd Normal form. Document your data acquisition methodology if applicable. A poster template will be provided.

3. **Class Presentation**: Present your project at the end-of-semester class conference. Attendance is *required* that week! Wellness Day requests are not accepted.

Guidelines

1. Your project must be implemented using MySQL. (If you need to use a different database, please justify your choice in your project proposal.)

2. Use primary data sources. Use of Kaggle datasets is not allowed! Document your data sources in your project proposal! Sometimes, students come up with a fantastic idea only to discover that the required data is not available. Verify the feasibility of your project as you develop your project proposal.

3. Your project deliverables will include:

    a. A 1-slide poster using the RISE poster template.
    b. A full-page PDF of your conceptual database design (ER model)
    c. Your DDL (Create table statements) in .SQL format
    d. A separate file containing documented .SQL queries

Grading

Project grading will consider many factors including:

1. The significance of your project
2. The extent to which your project demonstrates database design proficiency
3. Your conference presentation
4. The readability and clarity of your final poster.